

# Atlas of halving: what one refinement step does, and what emerges, in each dimension

**Purpose, 2026-09-27 (user direction: a collective understanding of the halving of the lattice and what emerges in each case and dimension, as the mid-term goal).** This note is the shared map. Each cell says what one halving does, what it leaves behind, and what survives the limit, with the status of the statement and the note that holds it. It adds no new theorem; it fixes one vocabulary so that the agents working here (Claude, GPT-6 Astra, Fable reviewers) and the user fill the same cells. Open cells are listed in § 5.

Status labels: **proved** (theorem in a repository note, reviewed or not as marked), **exact** (closed-form identity), **formal** (derivation assuming an unproved estimate), **known** (established literature, cited), **open**.

## 1. The operation

**Halving one direction**  $a_1 \mapsto a_1/2$  of a lattice with plaquette heat times  $t_{\mu\nu} = \lambda_D a_\mu a_\nu / \prod_{\rho \neq \mu, \nu} a_\rho$  (series/parallel note, eq. (1), small-field regime) splits exactly into two moves.

- **Series move.** A face containing the refined direction is cut in two; its heat time halves. Integrating the cutting edge is a convolution of the two half-weights: character coefficients multiply. For heat-kernel weights it closes exactly in every dimension.
- **Parallel move.** A face transverse to the refined direction gets a new copy in the mid-plane, both at doubled heat time. Integrating the mid-plane leaves the factor  $\Psi$ : a  $(D-1)$ -dimensional gauge theory whose edges carry Brownian-bridge laws fixed by the coarse field. Pointwise products of weights add their logarithms; the heat kernel is closed under this only up to vortex terms (zero-spacing note, Proposition 1).
- **Count.** A halving produces parallel moves in  $\binom{D-1}{2}$  transverse planes. An isotropic step is  $D$  directional halvings; the isotropic heat time scales as  $t \propto \lambda_D a^{4-D}$ .
- **Time-only halving** (Euclidean time at fixed spatial lattice) sends the transverse heat times to infinity; the Hamiltonian limit needs exponential (Wilson-type) magnetic weights (zero-spacing note, § 1).

**Newton's halving** is the one-dimensional case: inserting a time  $t_{2.6}$  in a cell is a series move (refinement note, § 2). A physical record at the inserted time is a new variable attached to the body, the analogue of a parallel insertion. Forgetting a Gaussian record is an exact parallel move with defect proportional to  $\hbar^2$ , the identity for  $\hbar = 0$  (record as parallel move).

## 1b. Cuts at any position (user remarks, 2026-09-27/28)

Halving is the easiest cut; a cut at any fraction  $s \in (0, 1)$  of a cell performs the same two moves, with  $s$  as a parameter.

- **Series move.** The cut face splits into heat times  $st$  and  $(1-s)t$ ; the convolution returns  $t$  exactly, for heat-kernel weights, in every dimension.

- **Parallel move.** With trapezoid (dual-length) face weights every transverse face keeps heat time  $2t$ , and the inserted edges carry the bridge law at fraction  $s$ , variance  $s(1-s)t_e$ . The free-field defect scales exactly by  $4s(1-s)$  (Corollary 2<sub>s</sub>), and the  $U(1)$  Theorem 5 holds for every  $s$  (Corollary 5<sub>s</sub> there): halving is the largest single step, and an off-centre cut is gentler and shrinks the mesh less.
- **Newton.** One cut spends exactly  $3s(1-s)K_\tau$  of the cell action  $K_\tau = F^2\tau^3/(24m)$ , the Cameron–Martin energy of the Schauder hat it inserts; the shares add over any cut sequence and exhaust  $K_\tau$  iff the mesh vanishes (cut-measure note). The Galileo action is an additive measure on the cut process, with the nested additivity of the Lévy–Ciesielski construction of the Brownian bridge. With a mark floor  $\kappa$ , marks at any finite set of cuts distinguish the force with squared statistical distance exactly the spent action over  $\kappa$  (Proposition 7 there), so the cut measure and the record measure coincide.
- **Windings and images.** For  $U(1)$  a cut’s bridge law is a positive winding mixture: its weights are  $s$ -independent and the shifts  $2\pi(1-s)W$  form  $\mathbb{Z}_2$  only at halving and  $\mathbb{Z}_q$  at  $s = p/q$ . For every compact simply connected group the midpoint expectation of every character is exact, a sum over weights of coroot image sums (centre note, Theorems 1 and 1’). Its signed Weyl-polynomial amplitudes describe relative Cartan images; positivity of a mixture and homotopy winding sectors in the group are additional interpretations unsupported by that expansion.
- **Which part of the centre a cut reaches.** A cut at  $p/q$  reaches exactly the subgroup  $\mathbb{Z}_{\gcd(q,N)}$  of the centre of  $SU(N)$  (its Corollary 3). So the  $SU(2)$  halving carries Dirac’s belt trick, the image  $n$  acting by  $(-1)^n$  on half-integer spins, while dyadic refinement reaches only its identity for  $SU(3)$  and a trisection carries the triality. Flipping one mid-edge by a central element inserts thin centre flux into its incident faces. Open: whether this ties refinement to centre vortices, and whether triadic refinement suits  $SU(3)$  better. Astra’s Round 5B Part A referee is complete (2026-09-28): the exact identities and torsion criterion are accepted, with scope corrections.
- **The rod.** The stick of Zhuangzi 33, 一尺之捶，日取其半，萬世不竭, halved only in its remaining piece, spends  $\frac{6}{7}K_\tau$  and converges to a point, the Mohist 端 of Canon B; refinement, which spends all of  $K_\tau$ , cuts every piece again (cut-measure note, § 3). The parallel with Zeno’s dichotomy is recorded as convergence.
- **The frame.** A cut at an arbitrary, even irrational, position is the geometers’ cut of Book I’s closing scholium. The questions for each cell become: does the limit depend on the cut sequence (for Newton, no, proved in the cut-measure note; for the free field, expected, a convergence question for the discrete-exterior-calculus Hodge star that the trapezoid weights are, see Corollary 2<sub>s</sub>), and does the universal part survive when  $s$  varies. For cell 4 it does, conditionally: for shape-regular schedules under the matching hypotheses of the four-dimensional note (its § 7), the coefficient per logarithm of the physical scale is  $2b_0$  whatever the cut positions. An earlier guess here,  $\log(1/s)$  per cut, follows one daughter only and omits the other’s dual-volume weight.

## 2. The atlas by dimension

| $1 + n$           | Series move                                                                                                 | Parallel planes per halving | What one step leaves                                                                                                                                                                     | What survives the limit                                                                                                           | Status and notes                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| $0 + 0$           | none (a single weight)                                                                                      | none                        | the weight $k_t(U)$ itself; a single integral with an $e^{-c/\hbar}$ structure                                                                                                           | a large- $N$ transition (Gross–Witten)                                                                                            | known; dimension ladder                                                  |
| $1 + 0$<br>Newton | exact after one cubic counterterm<br>$-F^2\hbar^3/(24M)$ ;<br>the defect $-F^2uv\hbar/(8M)$ is a coboundary | none                        | a scalar                                                                                                                                                                                 | unobserved: the exact propagator at every $\hbar$ ; recorded: a floor only if joint determinacy fails                             | proved; refinement note, fifth postulate                                 |
| $1 + 1$           | exact (heat-kernel convolution)                                                                             | 0                           | nothing for heat kernels; for any action, $a^{-2}(1 - \hat{c}_R)$                                                                                                                        | a conjugation-invariant Lévy exponent $\psi(R)$ ; Yang–Mills iff Lindeberg; circle spectrum $\hbar cL\psi(R)$ ; no local particle | proved (Theorem 2 of the zero-spacing note); known (Lévy)                |
| $1 + 2$           | exact                                                                                                       | 1                           | free: an irrelevant quadratic form of relative size $a^2K^2/16 + (Ga)^2/8$ ; $U(1)$ : vortices of density $e^{-\pi^2/(8t)}$ ; $SU(N)$ : coupling shifts $O(t)$ , curvature term explicit | free photon for $U(1)$ at fixed $\lambda_3$ (every lattice gap dies); conjectured gap $C_3\hbar c\lambda_3$ for $SU(N)$           | proved for free and $U(1)$ ; formal for $SU(N)$ (Proposition 7); see § 3 |

| $1 + n$           | Series move | Parallel planes per halving | What one step leaves                                                                                        | What survives the limit                                                                                                                                                                        | Status and notes                                                                           |
|-------------------|-------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| $1 + 3$           | exact       | 3                           | $t = g^2$ fixed: coupling shifts that recur each step; large fields suppressed only as $(a\Lambda)^{2b_0c}$ | running coupling, $\Lambda = \mu e^{-1/(2b_0g^2)}$ ; small instantons die $(11N/3 > 4)$ ; dislocations need $c > 2/b_0$ ( $\frac{6}{11}$ of an instanton for $SU(2)$ ); $U(1)$ : Coulomb phase | known (running, 6/11 criterion, Driver); the measure-level route fails for $U(1)$ (proved) |
| $1 + n, n \geq 4$ | exact       | $\binom{n}{2}$              | $t$ grows as $a \rightarrow 0$                                                                              | no small-field regime; a phase transition, no established continuum limit                                                                                                                      | known (Creutz for $SU(2)$ in 5D)                                                           |

The same heat time  $t = \hbar g_{\text{cl}}^2 a^{4-D}$  controls the column “what one step leaves” . For  $D < 4$  refining and taking  $\hbar \rightarrow 0$  push it the same way; in mechanics refining makes each cell more quantum (dimension ladder, § 2).

### 3. The atlas for $1 + 2$ by group

| Group                   | Isolated cube | Full mid-plane                                           | One step, measure level | Iteration and limit                                                                                 |
|-------------------------|---------------|----------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------|
| free ( $\mathbb{R}^n$ ) | exact         | exact defect (Proposition 2 of the series/parallel note) | Gaussian, exact         | blocked actions converge to a Gaussian fixed point (known, Bell–Wilson; for this blocking asserted) |

| Group   | Isolated cube                                                                                                                                                | Full mid-plane                                                                                                                                                    | One step, measure level                                                                                                                                                                                 | Iteration and limit                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| $U(1)$  | exact<br>(Proposition 3)                                                                                                                                     | exact charge form, vortex bound<br>(Proposition 4, Corollary)                                                                                                     | one step equals the free step up to density $e^{-\pi^2/(8t)}$<br>(Theorem 5, refereed);<br>whole measure within TV $2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}$<br>of the monopole-free part<br>(monopole note) | free photon<br>(Gross 1983, known)                             |
| $SU(2)$ | midpoint characters exact for every spin; spin- $\frac{1}{2}$ and spin-1 cube sectors exact, with matrix image bounds for spin 1 (closed form, Theorems 1–4) | formal order- $t$ expansion: cut shifts positive, transverse negative; other local operators have dimension at least six; explicit torus winding term             | formal;<br>one-step normalized small-field bound against a covariant reference proved (small-field § § 13–14); iteration open                                                                           | open                                                           |
| $SU(3)$ | curvature term explicit, softening $t_j X_j ^2/512$ (group-general form of Proposition 6)                                                                    | small-field one-step normalized bounds transcribed from $SU(2)$ , $\mathbb{Z}_3$ sectors, bridge tail by Li–Yau (small-field § 16, unrefereed); large fields open | formal                                                                                                                                                                                                  | open; the gap $C_3\hbar c\lambda_3$ is the infrared obligation |

#### 4. What emerges, read across the atlas

Why ultraviolet halvings bear on an infrared gap is explained, for the paper, in the didactic note.

- **Series moves never produce anything new.** In every dimension they close exactly for heat-kernel weights, and in mechanics after one scalar counterterm. The one-dimensional Newton problem and the two-dimensional gauge theory are all series.

- **Parallel moves carry everything that renormalizes.** Their number,  $\binom{D-1}{2}$ , is zero exactly where the theory is exactly soluble.
- **Additivity is universal; locality is one-dimensional** (2026-09-28). Integrating out a Gaussian step leaves the Schur complement of the action (the Dirichlet principle), and successive Schur complements compose exactly ((13) of the four-dimensional note); in Newton's cell this is Theorem 2 of the cut-measure note. Only one dimension grants locality: the minimizer between two cut points is the chord, so the blocked action is again nearest-neighbour and the series move closes, as in  $1 + 1$ . With transverse planes the Schur complement is non-local, and its local truncation error is the parallel defect ((5) and Corollary 2<sub>s</sub> of the series/parallel note).
- **The exponential ladder.** Each compact correction left by a step is  $e^{-c/t}$  (vortices, monopoles, dislocations). In  $D < 4$  it is summable per physical volume at fixed coupling, so compact effects die; in  $D = 4$  the running turns it into a power of  $a$ , so survival is a threshold condition; in  $D > 4$  there is no small  $t$ . In  $1 + 3$  at one loop,  $a\Lambda = e^{-1/(2b_0t)}$  with  $t = \hbar g^2(a)$ , so  $e^{-c/t} = (a\Lambda)^{2b_0c}$  and  $2b_0c$  is the scaling dimension of the correction. This is the bookkeeping of 't Hooft's infrared renormalons, whose Borel-plane positions are fixed by  $b_0$  and an operator dimension; the per-volume threshold  $2b_0c = 4$  sits at the gluon-condensate renormalon ('t Hooft, Erice 1977, publ. 1979; metadata, renormalon positions quoted from memory). The generated gap is itself on the ladder:  $am \propto e^{-1/(2b_0\hbar g^2)}$  is the member with  $2b_0c = 1$ . The bridge image terms sit far above the per-volume threshold  $2b_0c = 4$ :  $2b_0c = 11N/3$  for root images ( $c = 8\pi^2$ ) and 33 for the  $SU(3)$  centre images of a trisection ( $c = 24\pi^2$ , centre note) (2026-09-28). These are formal powers from near-identity image exponents; normalized bounds uniform in boundary data and under iteration remain to prove.
- **What survives is trajectory-dependent.** Compact  $U(1)$  in  $1 + 2$  keeps a gap along  $\lambda_3 a \simeq c_0/(2 \log(1/a))$ ,  $c_0 \approx 4.99$  the monopole exponent, where the monopole density per physical volume diverges, and loses it at fixed  $\lambda_3$  (Göpfert–Mack versus Gross).
- **The action floor survives what removes the mass gap** (user observation, 2026-09-28). With  $t = \lambda_3 a = \hbar g_{\text{cl}}^2 a$  the compact corrections of a step,  $e^{-c/(\hbar g_{\text{cl}}^2 a)}$ , vanish as  $a \rightarrow 0$  at fixed coupling and as  $\hbar \rightarrow 0$  at fixed  $g_{\text{cl}}$  and  $a$ , while the  $w = 0$  Gaussian term of (9) of the series/parallel note, of width set by  $\hbar$ , remains. Both  $U(1)$  continuum limits on record are Gaussian: Gross's photon, with field-strength covariance  $\hbar g_{\text{cl}}^2 K_0$ , and Göpfert–Mack's free massive scalar, canonically normalized along a trajectory on which  $\lambda_3$  diverges. The refinement removes the lattice mass gap and keeps the unit of action, and  $\hbar$  was first measured in this gapless theory in  $1 + 3$ , cavity radiation (Planck's talk of 14 December 1900, written up as Ann. Phys. 309, 553 (1901), through the resonators' energy elements; the field-mode reading is Debye 1910, metadata; a finite cavity keeps the box gap  $2\pi\hbar c/L$ ). Two gaps of different kinds: a mass gap, generated and trajectory-dependent, and an action floor, supplied in  $e^{-S/\hbar}$ , that every cell of every limit keeps (each cut weighted by  $e^{-\text{share}/\hbar}$ , cut-measure Prop. 4). In G07's terms both are unit multiples of the fixed constants; the floor is supplied, while a generated gap needs two scale-free structures that do not commute (G07 § 1; G08).
- **Constants that emerge.** Mechanics: one scalar counterterm per cell, and one action constant if joint determinacy is denied (Theorems B and B' of the fifth-postulate note).  $1 + 1$ : the Lévy exponent (one coupling for Yang–Mills).  $1 + 2$ : one coupling, shifted summably.  $1 + 3$ : one scale  $\Lambda$  by dimensional transmutation.

## 5. Open cells

1.  $SU(2)$  cube: filled through  $J = 1$  (Theorem 4, exact diagonal entries, image bounds and cube sector);  $J > 1$  entries and control of the full spin sum remain open.
2.  $SU(2)$  full mid-plane in  $1 + 2$ : order- $t$  calculation refereed as formal; small-field note § 12–14, Rounds 14–16, refereed by Claude: a covariant Gaussian reference with 't Hooft flux sectors, and normalized weak  $P(1/2 - \delta)$  against it for one barriered step, uniformly in plane size, for equal and unequal boundary layers with the cut faces budgeted (saddle determinant, Laplace remainder and amplitudes included; remainder densities face-local in size). Next: locality of the remainder's dependence on the field (soft-barrier realization, boundary scores, barrier cost; § 14.2), then images/large fields, perturbed-action stability and iteration; the fixed-frame transport-value comparison remains open; Proposition 4's rejection stands.
3. Stability of Hypothesis  $P(\alpha)$  under iteration. For  $U(1)$  in  $1 + 2$ , reduced to exact Gaussian blocking (Corollary 2' of the monopole note); open for  $SU(N)$ .
4.  $1 + 3$ : composition, Thms 1–4 refereed by Claude; § 7 gives generated cubic/quartic kernels, decay, finite one-step subtracted matching with explicit constants and uniform iteration hypotheses (Round 10, formal, unrefereed); G1 refereed. Open: its (14), uniform subtracted cubic/quartic contractions; sublinear endpoint matching and remainder sum give the average  $-2b_0 \log 2$  (conditional Theorem 4; one-step shifts refereed).
5. Time-only halving. Structural reading, 2026-09-27, of known results. At fixed spatial lattice the electric faces make the series moves: heat-kernel electric weights are a semigroup in the time step, so they compose exactly. The magnetic faces make the parallel moves: with exponential weights  $e^{-a_0 V}$  they close exactly in their own family, since products of such weights add the exponents. The whole defect of a time halving is the non-commutation of the two families, and it vanishes as  $a_0 \rightarrow 0$  by the Trotter product formula (Trotter 1959; Chernoff 1968; metadata; strong convergence for the Laplacian on  $G^E$  plus a bounded magnetic potential). This is why the Hamiltonian limit (Kogut and Susskind 1975; transfer matrix: Creutz 1977; metadata) exists at every fixed spatial lattice, while spatial halving leaves the non-closing factor  $\Psi$ . Open: an operator-norm rate, which needs domain estimates for the commutator of the Laplacian with  $V$ .
6. Filled 2026-09-27 for Gaussian records (record as parallel move): forgetting a record convolves momentum with variance  $\hbar^2/(4\sigma^2)$ ; precisions add in parallel; refinement converges iff  $\sum \sigma_j^{-2} < \infty$ ; identity for  $\hbar = 0$ .
7. The infrared: from ultraviolet control in  $1 + 2$  to  $C_3 > 0$ .
8. Cuts at any position (§ 1b), open rows: the free-field independence of the limit from the cut sequence (a convergence question for the discrete-exterior-calculus Hodge star on nested tensor grids); for  $SU(3)$ , whether triadic refinement, the only kind whose image terms reach the centre (centre note), organizes the large-field terms better than dyadic refinement.

## 6. Consequence for STATE

STATE carries the mid-term goal and points here. Each result that fills a cell updates the cell in this note and nothing else; the notes that prove results remain the record.